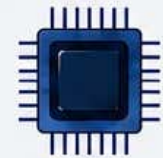
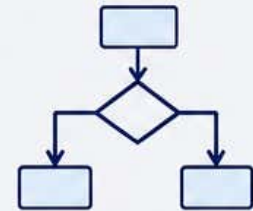




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Syllabus



VSC: Laboratory on Laboratory on R Programming

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112288	Laboratory on R Programming	2	60	4Hrs.	50

Course Objectives:	1. To understand the fundamentals of R Programming for data analysis. 2. To develop skills in data manipulation, visualization, and statistical analysis using R. 3. To apply R libraries and functions for solving real-world problems. 4. To build analytical and programming skills using R tools and techniques.						
Course Outcomes:	On completion of the following syllabus the students will be able to - 1. Identify and use basic R programming syntax and tools. 2. Apply data handling and visualization techniques using R. 3. Analyze datasets using statistical methods in R. 4. Develop simple analytical solutions and mini projects using R programming.						
Contents				Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies	
List of Practical: 1. Installation and configuration of R Programming and RStudio environment for statistical computing. 2. Write an R program to perform basic arithmetic, relational, and logical operations. 3. Create and manipulate vectors, matrices, arrays, and lists in R programming. 4. Write an R program to perform operations on data frames and factors. 5. Import and export CSV and Excel datasets using R functions. 6. Perform data cleaning and handling of missing values using R. 7. Implement string manipulation functions and date-time operations in R. 8. Write R programs using conditional statements and looping constructs. 9. Create user-defined functions in R for mathematical and statistical operations. 10. Perform sorting, filtering, and merging of datasets using R programming. 11. Create bar charts, pie charts, and histograms using R visualization libraries. 12. Plot scatter plots, line graphs, and box plots for data analysis in R. 13. Calculate Mean, Median, Mode, Variance, and Standard Deviation using R. 14. Perform correlation and covariance analysis on datasets using R.						Use any pedagogical technique suitable for the topic	

	15. Implement simple linear regression analysis using R programming. 16. Perform hypothesis testing using statistical functions in R. 17. Use the ggplot2 library for advanced data visualization in R. 18. Analyze a real-world dataset using descriptive statistics and graphical representation in R. 19. Develop a mini project for student performance analysis using R programming. 20. Develop a mini project for sales prediction or survey data analysis using R tools and visualization techniques.			
References:	Weblink to Equivalent MOOC on SWAYAM if relevant: 1. https://www.youtube.com/watch?v=_V8eKsto3Ug 2. https://www.youtube.com/watch?v=iROHLA_TXQM 3. https://www.youtube.com/watch?v=h29g21z0a68 4. https://www.r-project.org/ 5. https://posit.co/products/open-source/rstudio 6. https://cran.r-project.org/ 7. https://cran.r-project.org/manuals.html 8. https://www.datacamp.com/tracks/r-programmer 9. https://www.coursera.org/learn/r-programming 10. https://www.kaggle.com/learn			

Assessment Rubric: Internal Examination Practical Course [(VSC)]

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Practical	Course Code: 112288	Laboratory on R Programming	Max Marks: 50
Sr. No.	Assessment Criteria		Marks
1.	Single Program Writing, Algorithm, Flowchart, Execution, with Sample inputs and Outputs, and Printout		20
2.	Record book preparation (Preferably handwritten) / Assignments preparation during semester		10
3.	Attendance in practical session / Students' overall performance		10
4.	Internal Viva Voce		10